

Domanda 18
Risposta
corretta
Punteggio
ottenuto 1,00
su 1,00

Given two random variables X and Y , Bayes Theorem implies that $p(y|x) = \frac{p(x|y)p(y)}{p(x)}$ where:

Scegli un'alternativa:

- ☐ a. $p(x)$ is called likelihood on x .
- ☒ b. $p(x)$ is called prior distribution on x . ✓
- ☐ c. $p(x)$ is called posterior distribution on x .

La risposta corretta è: $p(x)$ is called prior distribution on x .

Domanda 19
Risposta
corretta
Punteggio
ottenuto 1,00
su 1,00

In Maximum Likelihood Estimation, the prediction function is:

Scegli un'alternativa:

- ☒ a. A probabilistic function. ✓
- ☐ b. None of the above.
- ☐ c. A deterministic function.

La risposta corretta è: A probabilistic function.

Domanda 20
Risposta
corretta
Punteggio
ottenuto 1,00
su 1,00

Suppose a set of data $(x_i, y_i), i = 1, \dots, N, y_i = f(x_i) + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. In linear regression, the predictor function f is chosen as:

Scegli un'alternativa:

- ☒ a. $f(x, \theta) = \theta^T x$. ✓
- ☐ b. None of the above.
- ☐ c. $f(x, \theta) = \frac{\theta^T}{x}$.

La risposta corretta è: $f(x, \theta) = \theta^T x$.

Domanda 21
Risposta
corretta
Punteggio
ottenuto 1,00
su 1,00

Given a matrix $A \in \mathbb{R}^{m \times n}, m > n$, with $r = rank(A)$, then:

Scegli un'alternativa:

- ☐ a. None of the above.
- ☐ b. It is always possible to write A as $U\Sigma V^T$, where $\Sigma \in \mathbb{R}^{n \times n}$ is diagonal, $U \in \mathbb{R}^{m \times n}, V \in \mathbb{R}^{m \times n}$ are orthogonal.
- ☒ c. It is always possible to write A as $U\Sigma V^T$, where $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal, $U \in \mathbb{R}^{m \times m}, V \in \mathbb{R}^{n \times n}$ are orthogonal. ✓

La risposta corretta è: It is always possible to write A as $U\Sigma V^T$, where $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal, $U \in \mathbb{R}^{m \times m}, V \in \mathbb{R}^{n \times n}$ are orthogonal.

Sezione precedente

Vai a...